Memo

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Data Quality

We combined 1,067,371 transaction records and removed duplicate, invalid and non-product entries. We also adjusted purchases for matched returns. The final training data covers 5,463 customers and 3,684 products. We kept the last two months separate for testing, so future purchases were not used to build the recommendations.

Recommendation Results

We tested all methods on the same 1,906 customers. A recommendation counted as successful if at least one of its ten products appeared in the customer's held-out purchases. Our main SVD model succeeded for 23.50% of customers, compared with 17.37% for best sellers, an improvement of 6.14 percentage points. PCA reached 22.67%. In the purchase-value comparison, binary reached 24.50%, while raw quantity reached 14.27%.

Risks and Next Steps

Return matching is approximate because the data does not directly link every return to its original purchase. The evaluation covers existing customers and products with enough purchase history, so it does not measure performance for new customers or rare products. It also measures future purchases, not whether recommendations caused extra sales. The next step is a small controlled trial to check whether the recommendations improve sales before wider use.

Method

We chose Truncated SVD as our main method because our customer-product matrix is very sparse. Most customers only bought a small number of products compared to all available products. We also compared it with PCA. The results showed that SVD was faster, while both methods had similar reconstruction errors.

Storage and Speed

With 100 components, SVD took 0.2035 seconds to train, compared to 0.3140 seconds for PCA. The stored customer and product factors only needed about 6.98 MiB, and generating recommendations for all 5,463 customers took about 1.09 seconds. These results show that the model can reduce storage and generate recommendations quickly in our experiment.

Customer Segments

We also divided 5,463 customers into five groups based on their purchasing patterns. Some groups showed different preferences, such as home decorations, baking products, and lunch bags. We found that about 93.2% of customers were from the United Kingdom, so the country information did not help much in separating the groups.